Pledge: I pledge on my honour that I have neither taken nor given any help on this project – Abhijeet Mali.

Title: Computation of Stress Intensity Factors and T-Stress from Displacement Data and finite element simulation

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Abstract

This project studies how cracks behave in a solid body under plane strain using finite element simulations and displacement data near the crack tip. The main task is to calculate the stress intensity factors $K_I$and K_{II}, along with the T-stress, from the given displacement fields. These values are then used to rebuild the displacement fields, which are compared with the provided data to check accuracy. The stress patterns near the crack tip are also examined to see if they follow the square-root singularity predicted by Williams' solution, with reasons discussed for any differences. Finally, finite element simulations are carried out on standard geometries from ASTM D5045 (2022), and the crack tip parameters obtained from displacement fields are compared with those from load–displacement results. Overall, the study combines fracture mechanics theory with computational modelling to better understand crack tip behaviour and test the reliability of numerical methods.

Introduction:

Fracture Mechanics is the course which focuses on understanding how cracks initiate and propagate in materials. This behaviour of a crack under loading conditions is governed by parameters such as stress intensity factors (K_I, K_{II}) and T-stress, which describe the local stress and displacement fields near the crack tip. Accurate determination of these parameters is essential for predicting material failure and ensuring structural safety.

In this project, a cracked body under plane strain conditions is studied using finite element simulations. Displacement field data near the crack tip is provided, which serves as the basis for calculating the stress intensity factors and T-stress. The computed values are then used to reconstruct displacement fields, allowing comparison with the given data to check accuracy. The project also examines whether the stress and strain fields follow the square root singularity predicted by Williams' asymptotic solution.

Additionally, numerical simulations are carried out on standard geometries prescribed by ASTM D5045 (2022). These simulations enable comparison between crack tip parameters obtained from displacement fields and those derived from load–displacement relations. By combining theoretical fracture mechanics with computational modelling, the project aims to provide deeper insight into crack tip behaviour and evaluate the reliability of numerical methods in predicting fracture parameters.

Theory:

In linear elastic fracture mechanics, displacement field and stress field exact solution of isotropic and homogenous material given by Williams asymptotic series expansion. The stress intensity factors (K_I and K_{II}) characterize the singular stress fields near the crack tip, with their leading-order terms exhibiting a $r^{-1/2}$ dependence. For general problem considering mixed mode the displacement field can be written by superposition principal.

$$U = u(l) + u(ll)$$

$$U = f(KI, KII, T, r, \theta)$$

Where r and θ are polar coordinates.

T stress is the higher order, non-singular stress which influences the shape and size of plastic zone ahead of crack, it also correlates in-plane mean stress around the crack tip.

The ASTM D5045 standard specifies specimen geometry (3 point bend test) and loading conditions for fracture toughness testing of polymers and other materials. By simulating these geometries, one can compute displacement fields and fracture parameters numerically and compare them with experimental or load–displacement-based results. This ensures consistency between theoretical predictions, numerical simulations, and standardized testing procedures.

Methodology:

1. **Data reading and Processing:**
   - First displacement field data was extracted from a finite element simulation of a cracked body under plane strain conditions.
   - Then origin of coordinate system was set at the crack tip (x=0, $\theta = 0$)
   - Material properties used are E= 200 GPa & $\gamma = 0.33$
   - To ensure consistency with fracture mechanics formulations, all spatial coordinates and displacement values were converted into SI units.
   - The Cartesian coordinates were then transformed into polar coordinates (r, θ), where $r$denotes the radial distance from the crack tip and θrepresents the angular position measured from the crack line.

2. **Calculating SIF and T stress by using William's solutions:**
   - The near-tip displacement field in linear elastic fracture mechanics can be expressed using Williams' eigenfunction expansion. For an isotropic, homogeneous material under plane strain conditions, the displacement components are written as a series expansion in powers of $r^{n/2}$, with angular functions of θ.
   - In this study, the first two terms of the Williams expansion were considered, corresponding to:
     i. Mode, I stress intensity factor KI
     ii. Mode II stress intensity factor KII
     iii. The non-singular T-stress term
   - The displacement components $u_x$and $u_y$were expressed as linear combinations of these unknown coefficients. Rigid body motion terms, including translations and rotation, were also included in the formulation to eliminate artificial offsets in the displacement field.
   - The expression for mixed mode is given by eqn(1)

$$U = \sum_{n=0}^{\infty} \frac{r^{\frac{n}{2}}}{2\mu}\left\{a_n\left[\left(\kappa+\frac{n}{2}+(-1)^n\right)\cos\frac{n}{2}\theta-\frac{n}{2}\cos\left(\frac{n}{2}-2\right)\theta\right]\right.$$
$$\left.-b_n\left[\left(\kappa+\frac{n}{2}-(-1)^n\right)\sin\frac{n}{2}\theta-\frac{n}{2}\sin\left(\frac{n}{2}-2\right)\theta\right]\right\}+\alpha r\cos\theta+U_0$$
$$V = \sum_{n=0}^{\infty} \frac{r^{\frac{n}{2}}}{2\mu}\left\{a_n\left[\left(\kappa-\frac{n}{2}-(-1)^n\right)\sin\frac{n}{2}\theta+\frac{n}{2}\sin\left(\frac{n}{2}-2\right)\theta\right]\right.$$
$$\left.+b_n\left[\left(\kappa-\frac{n}{2}+(-1)^n\right)\cos\frac{n}{2}\theta+\frac{n}{2}\cos\left(\frac{n}{2}-2\right)\theta\right]\right\}+\alpha r\sin\theta+V_0$$

…….(1)

This overdetermined system was solved using the least squares method, yielding the expansion coefficients. The stress intensity factors were then obtained from the leading coefficients of the expansion using the relations:

$$\mathrm{K_I} = a_1\sqrt{2\pi}$$

$$\mathrm{K_{II}} = b_1\sqrt{2\pi}$$

The T-stress was obtained from the second-order term of the expansion. A convergence study was performed by increasing the number of terms in the Williams expansion to ensure that the extracted stress intensity factors became independent of the number of terms used.

3. **Displacement Field Reconstruction and Validation**

   - Using the extracted values of KI, KII, and T-stress, the near-tip displacement field was reconstructed from the Williams solution. The reconstructed displacements were evaluated along the crack line ($\theta \approx 0$)and compared directly with the FE displacement data.
   - Rigid body translation terms were added to the analytical displacement expressions to ensure proper alignment with the FE data. The comparison was carried out by plotting displacement as a function of radial distance r, allowing assessment of the accuracy of the Williams-based reconstruction.

4. **Stress Singularity and Boundary condition validation:**
   - The crack tip stress field was computed using the extracted stress intensity factors. In polar coordinates, the stress components $\sigma_{\theta\theta}$and $\sigma_{r\theta}$were evaluated along the crack line.
   - The variation of stress with radial distance was examined to verify the theoretical singular behaviour predicted by linear elastic fracture mechanics:

$$\sigma \propto r^{-1/2}$$

   - To confirm this behaviour, stress components were plotted against $r$on both linear and log–log scales. A reference line with slope $-1/2$was included in the log–log plots to demonstrate agreement with the expected singularity.

Results and Discussion:

1. **SIF and T stress**
   - SIF and T stress were calculated till 8 terms. As the number of terms increases accuracy of SIF also increases
   - Here plots generated from python are attached

N	KI	KII	T
8	1.648 MPa√m	0.098 MPa√m	0.110 MPa

Table 1: N, KI, KII and T for ***data1***

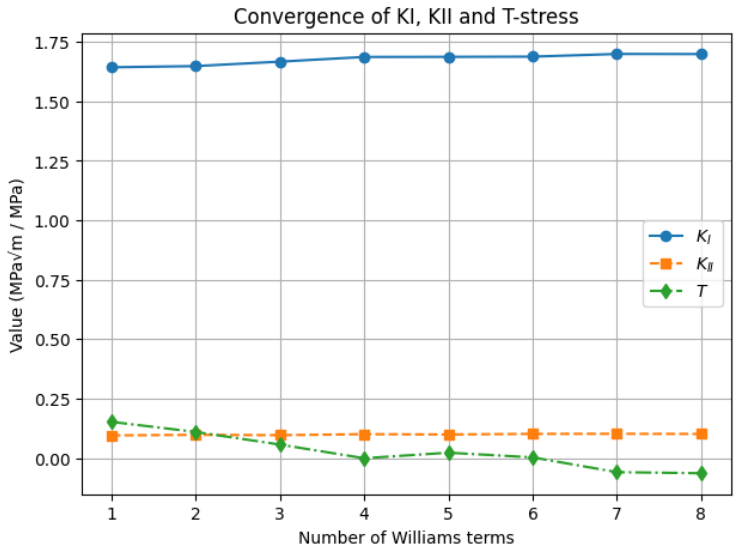

Fig 1: convergence plot for ***data 1***

N	KI	KII	T
8	1.757 MPa√m	2.373 MPa√m	1.252 MPa

Table 2: N, KI, KII and T for ***data2***

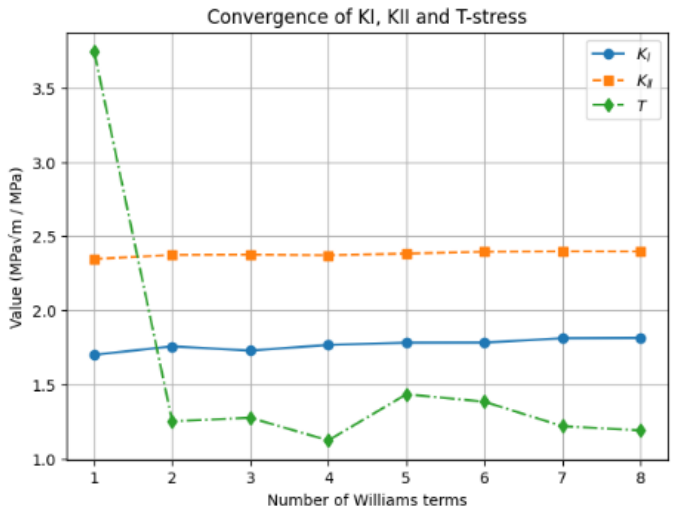

Fig 2: convergence plot for ***data 2***

N	KI	KII	T
8	-1.924 MPa√m	1.866 MPa√m	11.247 MPa

*Table 3: N, KI, KII and T for **data3***

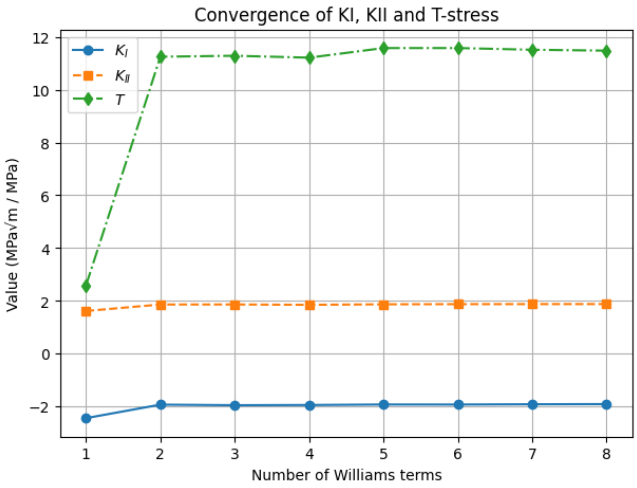

*Fig 3: convergence plot for **data 3***

2. **Displacement Field plot and comparison:**
   - Here we plotted and compared the displacement fields given in FEM excel data with that computed from code, i.e., using the solution for displacement from KI, KII, and T stress calculated earlier.
   - We calculated displacement fields at $\theta = 0$
   - Here plots generated from python are attached

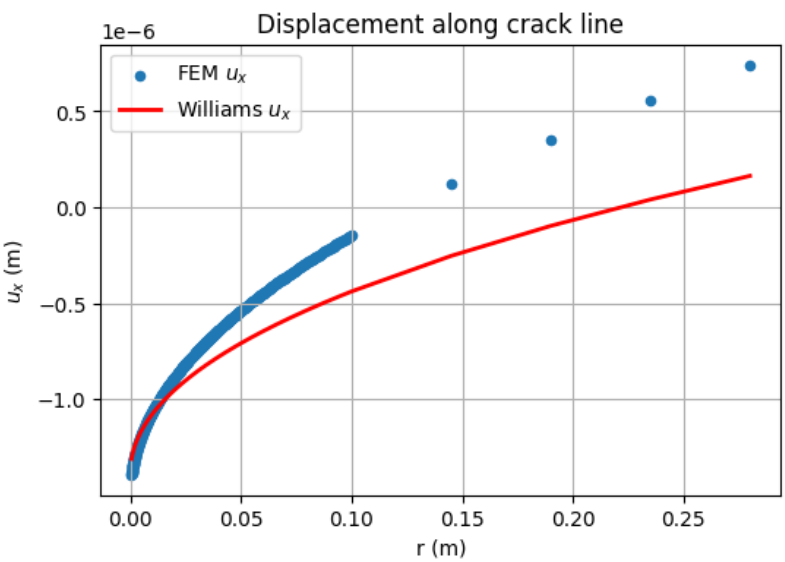

*Fig 4: Comparison plot of displacement fields for **data 1***

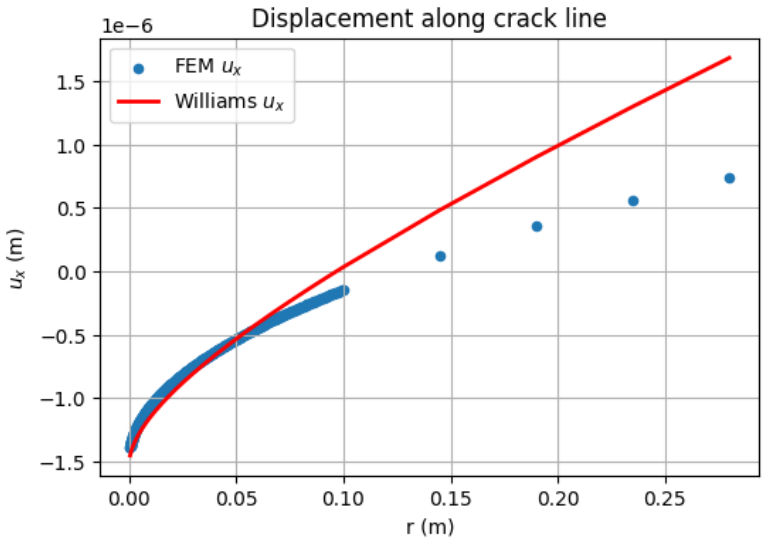

Fig 5: Comparison plot of displacement fields for ***data 2***

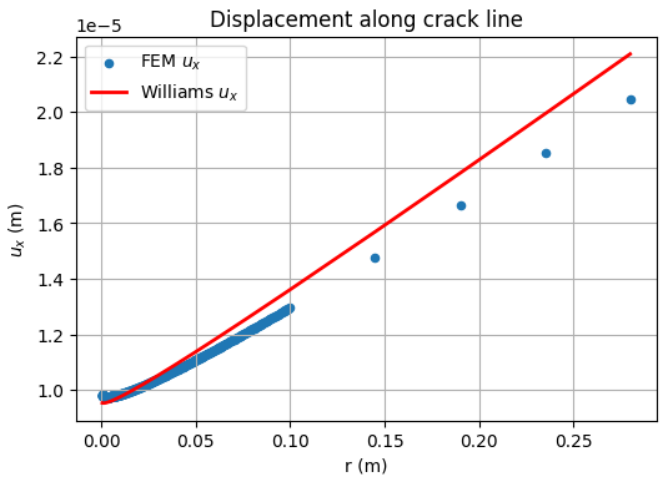

Fig 6: Comparison plot of displacement fields for ***data 3***

3. **Crack tip stress field singularity checks and boundary condition checks:**
   - Here we plotted $\sigma_{\theta\theta}$ with respect to r to check square root r singularity
   - For further confirmation we plotted $|\sigma_{\theta\theta}|$ vs r on log-log scale to get slope as a -0.5
   - Also, for additional check we plotted $\sigma_{r\theta}$ vs r to check Boundary condition

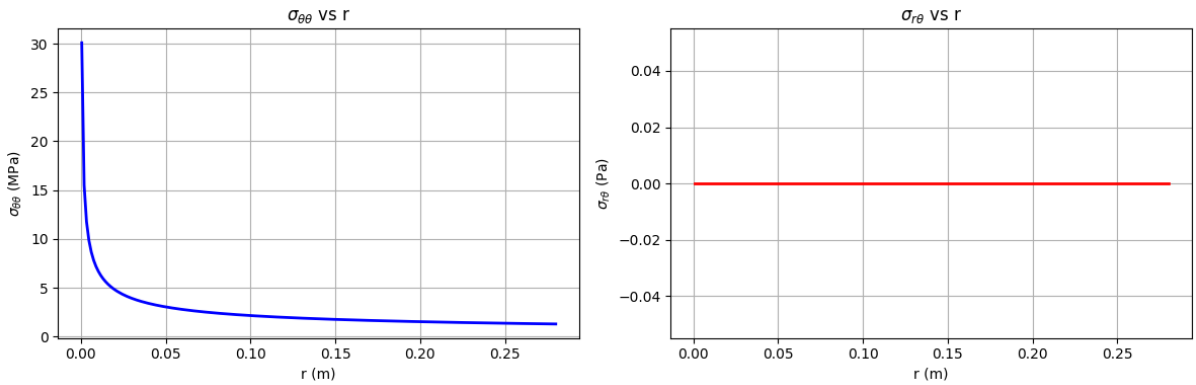

Fig 7: Plot of $\sigma_{\theta\theta}$ & $\sigma_{r\theta}$ vs r for ***data 1***

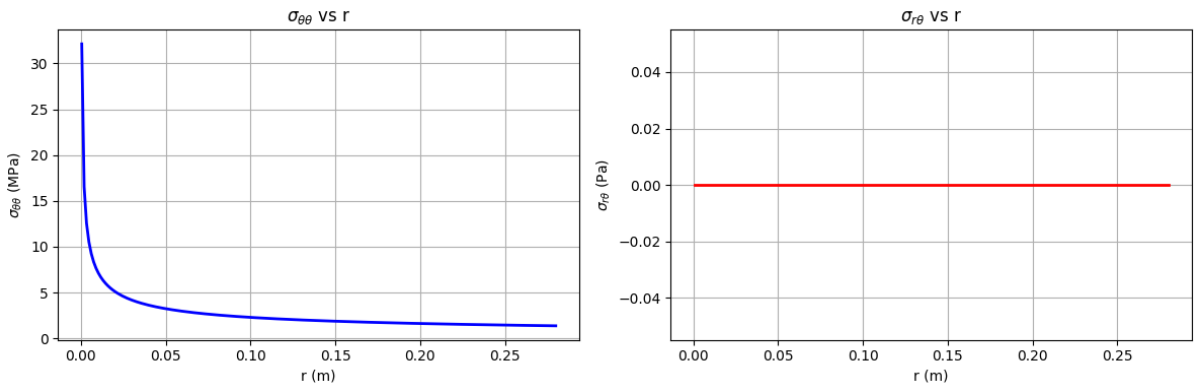

Fig 8: Plot of $\sigma_{\theta\theta}$ & $\sigma_{r\theta}$ vs r for ***data 2***

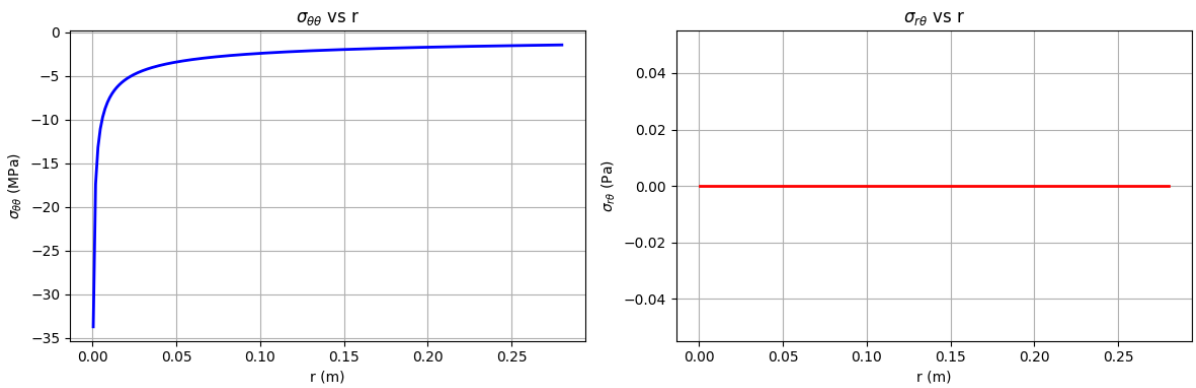

Fig 9: Plot of $\sigma_{\theta\theta}$ & $\sigma_{r\theta}$ vs r for ***data 3***

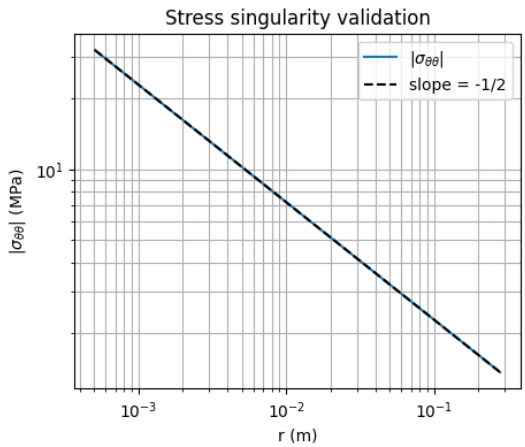

Fig 10: log-log plot of $|\sigma_{\theta\theta}|$ vs for ***data 1***

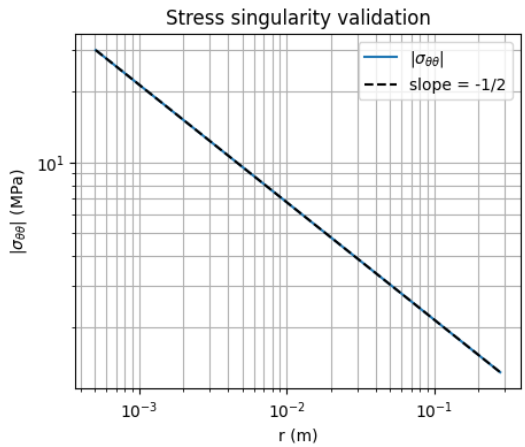

Fig 11: log-log plot of $|\sigma_{\theta\theta}|$ vs for ***data 2***

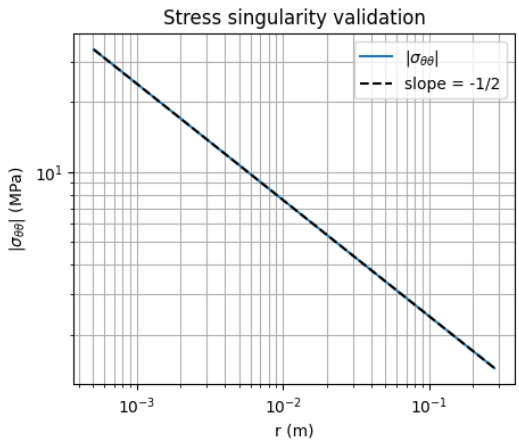

Fig 12: log-log plot of $|\sigma_{\theta\theta}|$ vs for ***data 3***

Finite Element simulation of ASTM D5045 (2022)

Methodology

Geometry and Specimen Configuration

A numerical simulation was carried out on a Single-Edge Notch Bend (SENB) specimen in accordance with ASTM D5045 specifications. The specimen dimensions were selected to ensure plane strain conditions at the crack tip.

- Specimen width (height), $W = 25.4$ mm
- Specimen length, $L = 111.76$ mm
- Specimen thickness, $B = 12.7$ mm
- crack length a = 12.7 mm

The span length Sbetween the support rollers was chosen as per ASTM D5045 recommendations.

Material Properties

The material was modeled as linear elastic, isotropic, and homogeneous. The following elastic properties were used:

- Young's modulus, $E = 200$ GPa
- Poisson's ratio, $\nu = 0.33$

Finite Element Model

The simulation was performed using Abaqus/Standard under a 2D plane strain assumption, which is appropriate given the specimen thickness and loading configuration.

Element Type and Mesh

- Element type: CPE8R (8-node biquadratic plane strain quadrilateral elements with reduced integration).
- To capture the crack tip singularity, the mesh was focused at the crack tip using a 'spider-web' configuration.

Boundary Conditions and Loading

The SENB specimen was subjected to three-point bending using rigid rollers:

- Two rigid support rollers were placed at the bottom of the specimen with a span S.
- One rigid loading roller was positioned at the mid-span on the top surface.
- A prescribed vertical displacement of 2 mm was applied to the top roller in the negative Y-direction, simulating a constant displacement-rate bending test.

Rigid rollers were modelled as analytical rigid bodies, and appropriate reference points were used to apply boundary conditions and loading.

Results

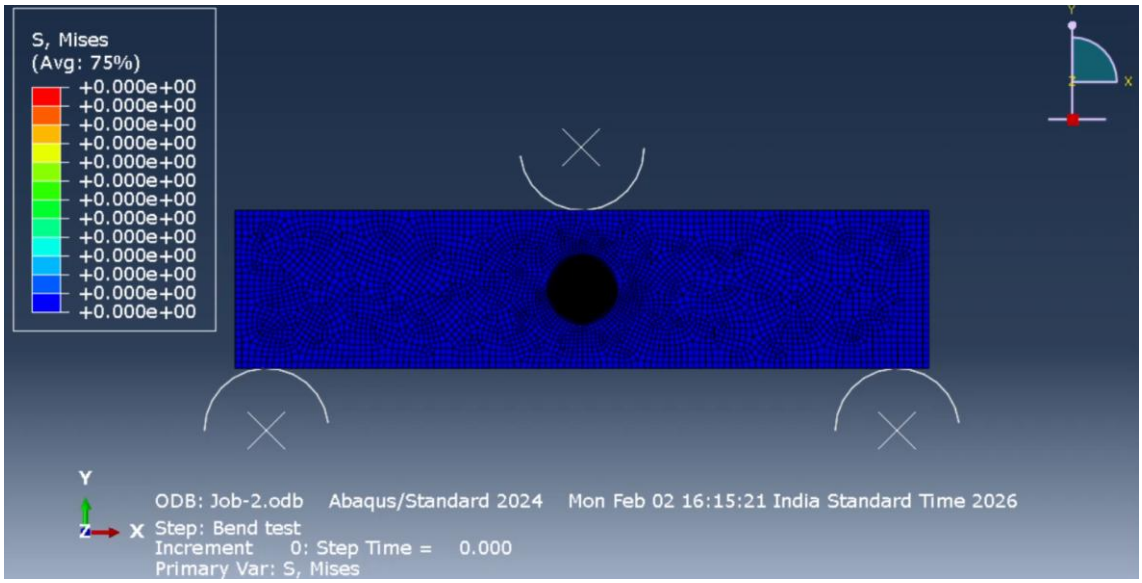

Fig 13: Test setup at the start of the simulation t=0 sec

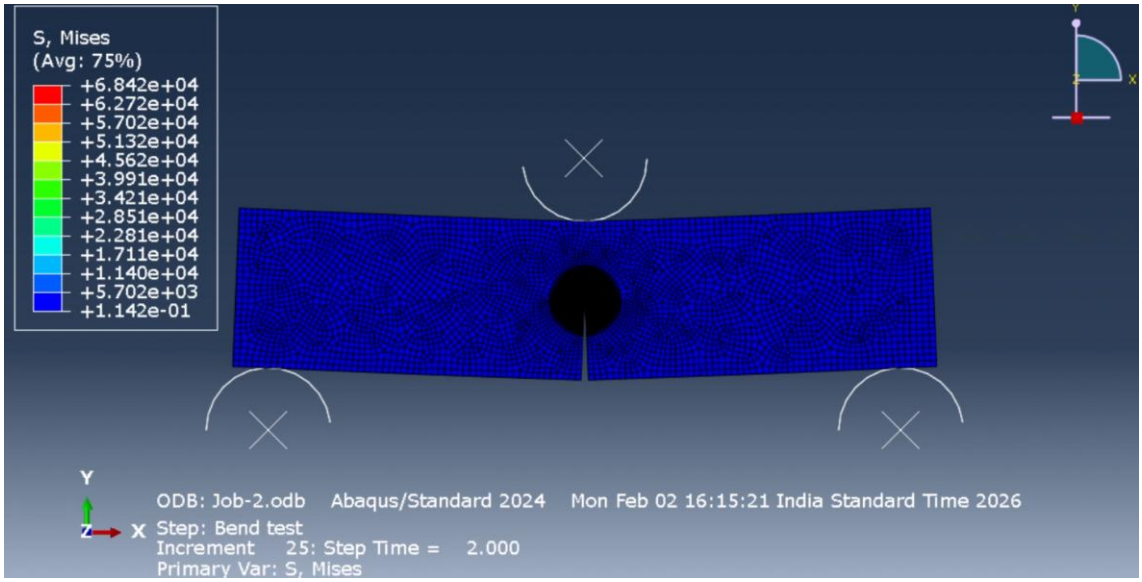

Fig 14: Test setup at the end of simulation t=2 sec

Figures 13 and 14 illustrate the undeformed (t=0 s) and deformed (t=2 s) configurations of the SENB specimen. The simulation was displacement-controlled, applying a uniform vertical displacement of 2 mm via the top rigid roller.

As observed in Figure 14, the specimen exhibits significant crack mouth opening displacement (CMOD) under the applied load. The stress distribution is highly localized, with the global mesh remaining in a low-stress state (indicated by the blue region) while high stress gradients develop near the crack tip singularity and the contact point. The symmetric deformation field confirms the validity of the boundary conditions and mesh alignment.

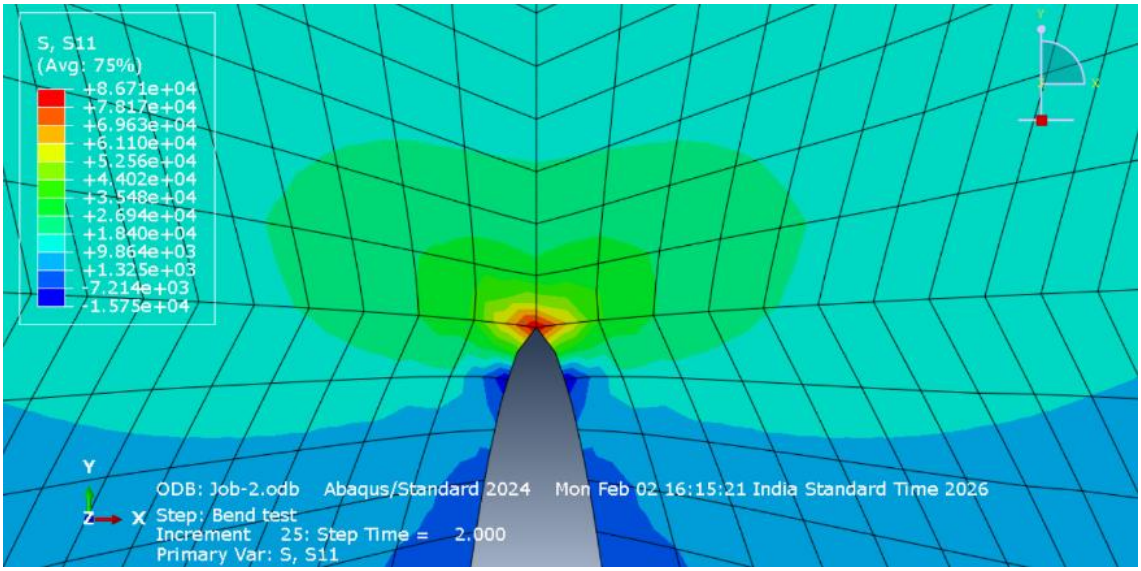

Fig 15: Sigma-11 stress state near crack tip at the end of simulation t=2 sec

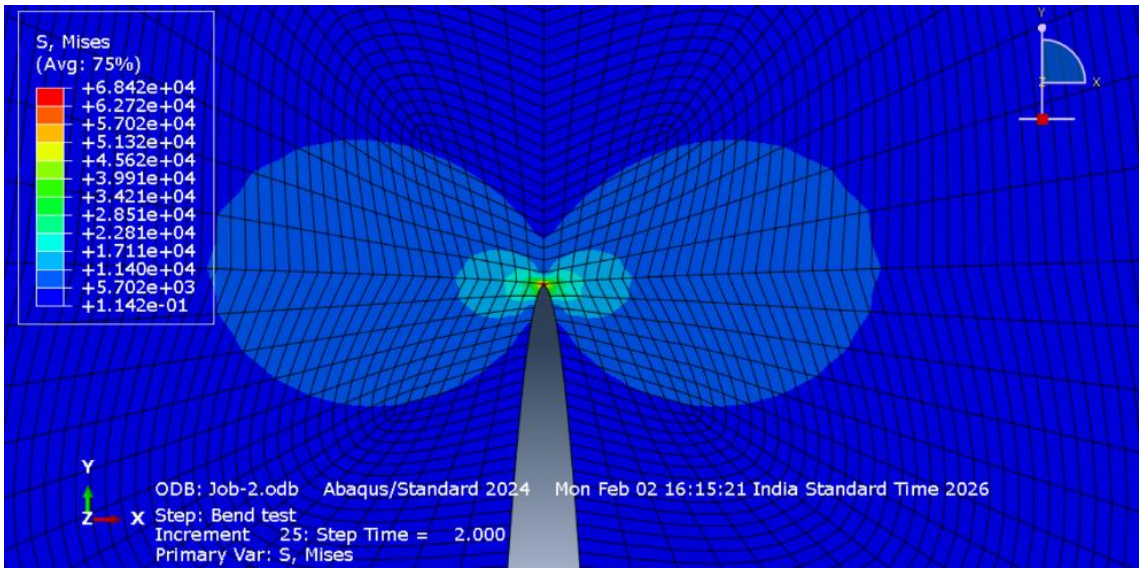

Fig 16: Von Mises stress field near crack tip at the end of simulation t=2 sec

Figure 15 displays the longitudinal stress component, which governs the Mode I opening. The contours show a symmetric tensile stress concentration ahead of the crack tip, consistent with the expected opening mode behavior.

Figure 16 illustrates the Von Mises equivalent stress. A characteristic 'butterfly' shape is visible in the stress contours, representing the zone where shear stresses are highest. The peak stress value recorded at the crack tip node is approximately 6.8e+04 MPa. It is important to note that this extreme value represents the mathematical stress singularity inherent in Linear Elastic Fracture Mechanics (LEFM), rather than a physical material response. As shown in the contours, the stress magnitude decays rapidly with distance from the tip, verifying the 1/sqrt{r} singularity field predicted by Williams' solution."

K1 and T-stress plot from simulation

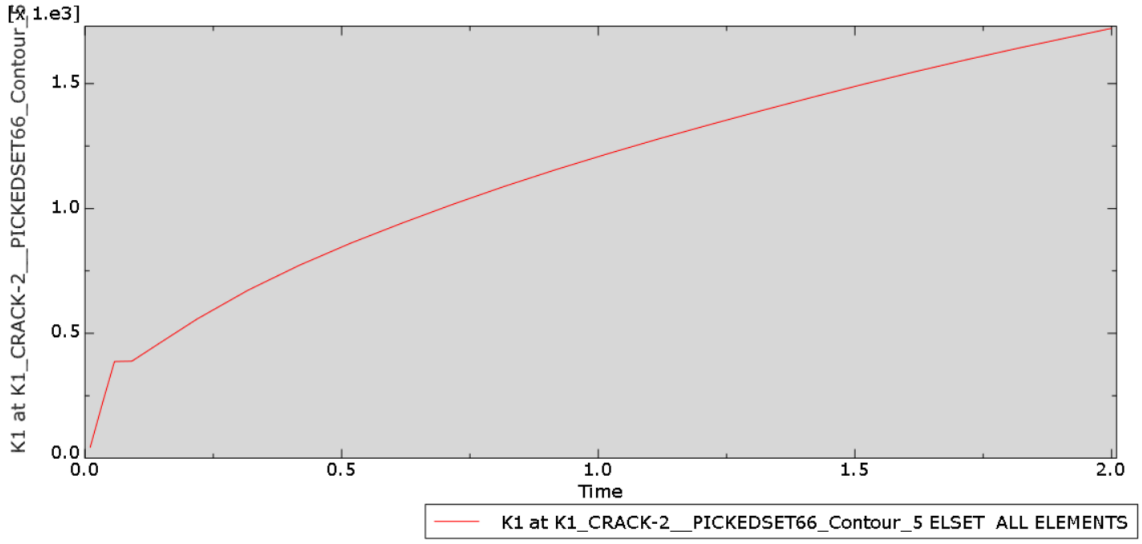

Fig 17: SIF K1 variation with time

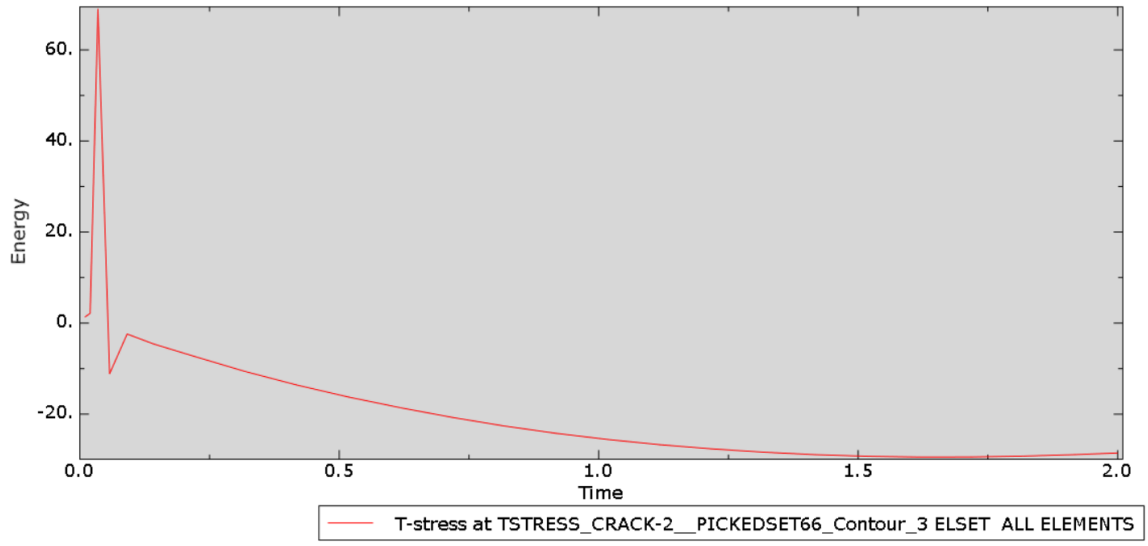

Fig 18: T-stress variation with time

Figure 17 presents the time history of the fracture parameters. The Stress Intensity Factor exhibits a increase commensurate with the applied load, reaching a peak value of 1733 MPa sqrt{m}.

The T-Stress (Figure 18) shows significant oscillation during the initial contact establishment ($t < 0.2$ s), which is an artifact of the contact stabilization algorithm. Following this transient phase, the T-Stress stabilizes to a steady compressive value of -26 MPa. The negative T-stress indicates high constraint and a stable crack path, suggesting the crack will tend to propagate straight ahead rather than branching.

Displacement field

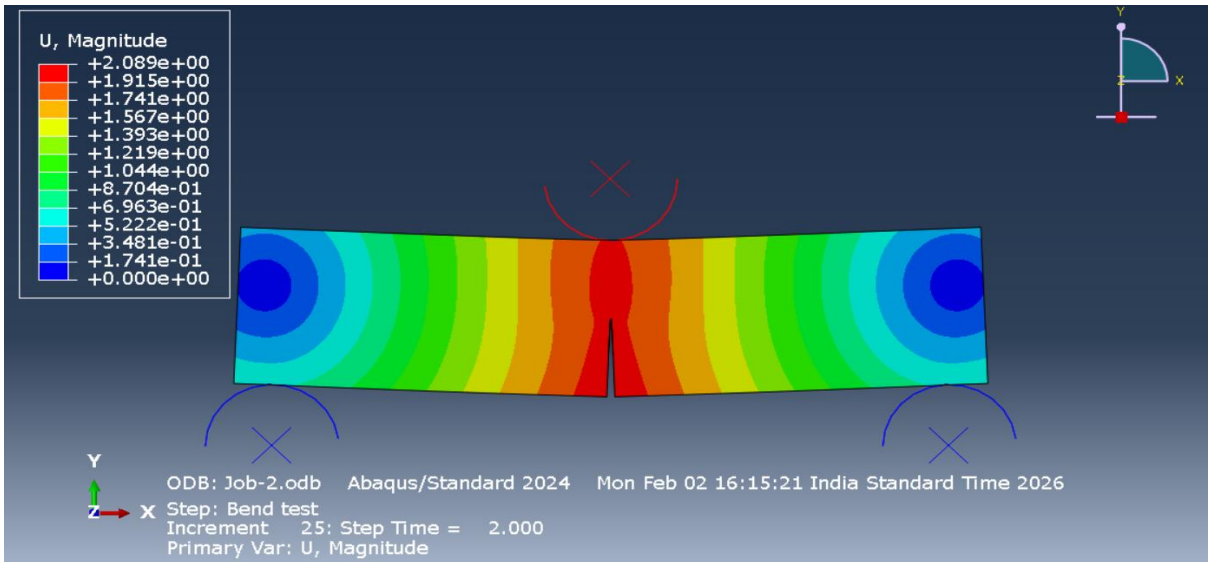

Fig 19: Displacement field at the end of simulation t=2 sec

The above figure illustrates the magnitude of the displacement field at maximum load. The contour plot reveals a symmetric deformation pattern characteristic of Mode I fracture, with maximum displacement occurring at the load application point. A clear crack opening profile is visible, verifying that the boundary conditions successfully constrained the specimen to in-plane bending without rigid body rotation.

Verification of Singularity: To validate the finite element model, the stress distribution near the crack tip was analysed. A log-log plot of the stress component versus distance from the crack tip is presented in below plot. The slope of the curve in the singularity dominated zone is approximately -0.5, consistent with the square root singularity predicted by Williams' asymptotic solution for Linear Elastic Fracture Mechanics (LEFM).

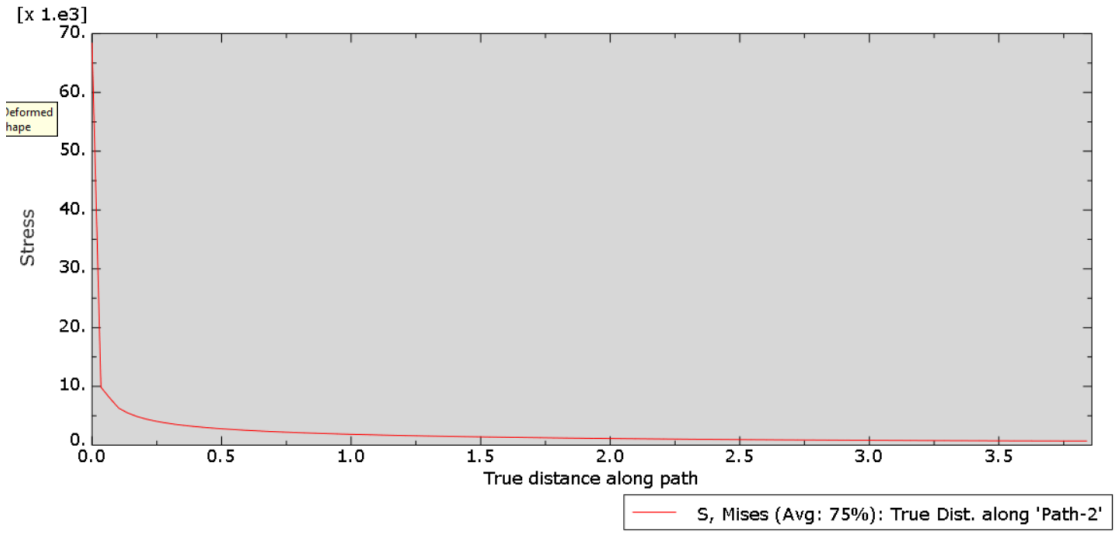

Fig 20: Von Mises stress field along a path ahead of the crack tip at theta=0

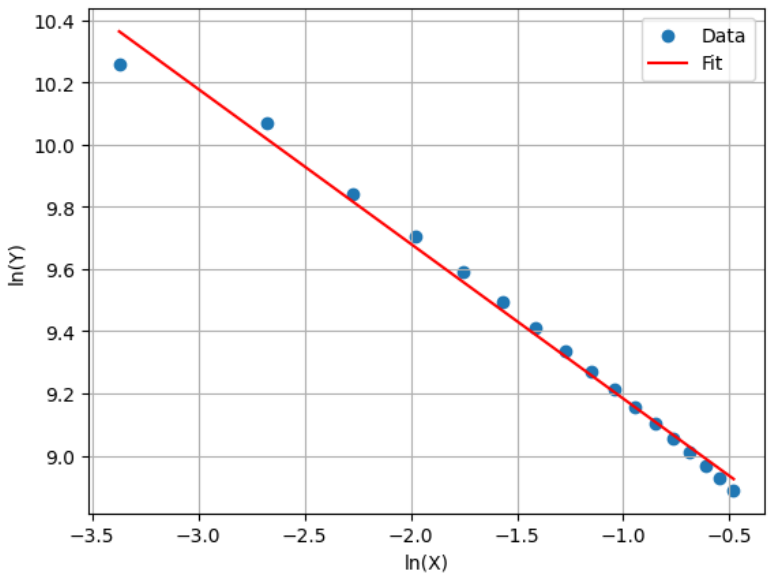

Fig 21: Log-log plot of stress vs distance for validating the 1/sqrt{r} relation

Stress Singularity: A path was defined along the uncracked ligament (theta=0) ahead of the tip. The log-log plot of Mises stress versus distance yields a slope of -0.4977, which is in excellent agreement with the theoretical singularity of -0.5.

Curve fitting using the equation,

$$y = cx^n$$

Gave the value of Exponent **n = -0.4977** & Constant **C = 5916.52**

Extraction of Fracture Parameters from DIC Displacement Fields for PMMA CT and 3PB Specimens

This part details the extraction and analysis of linear elastic fracture mechanics (LEFM) parameters from full-field displacement data obtained via Digital Image Correlation (DIC). Two polymethyl methacrylate (PMMA) fracture specimens-a 5 mm thick Compact Tension (CT) specimen and a 3-Point Bend (3PB) specimen-were subjected to Mode I loading. By fitting the experimental displacement fields to a generalized, multi-term Williams asymptotic expansion using an over-deterministic least-squares algorithm used in part A, the applied stress intensity factors and T-stress were computed. The reconstructed analytical fields showed good agreement with the experimental data, and the presence of the theoretical $r^{-1/2}$ stress singularity at the crack tip was successfully verified for both geometries.

2. Methodology

2.1 Data Acquisition and Coordinate Transformation

Raw images from the Mode I fracture tests were processed using the open-source DIC software Ncorr to extract the horizontal and vertical displacement fields. Because the Williams expansion inherently assumes the crack tip is located at the origin (0,0) with the crack plane perfectly aligned along the positive x-axis (theta = 0) a coordinate transformation was required. The origin was shifted to the physical crack tip, and rotational matrices were applied to the displacement vectors to mathematically align the crack plane with the x-axis.

2.2 Multi-Term Williams Expansion Fitting

The material properties for the PMMA sheets were defined as E = 2 GPa and v = 0.33. The problem was modeled under plane strain conditions. To compute the fracture parameters, the filtered experimental DIC displacement vectors {b} were mapped to the analytical Williams expansion matrix [A]. An overdeterministic linear least-squares was implemented in Python to solve for the coefficient vector {x}, which directly yielded KI, KII, & the T-stress.

3. Results and Discussion

3.1 Convergence of the Multi-Term Series

To accurately capture the near-tip fields without introducing numerical instability (overfitting), a convergence study was conducted to determine the optimal number of series terms (M).

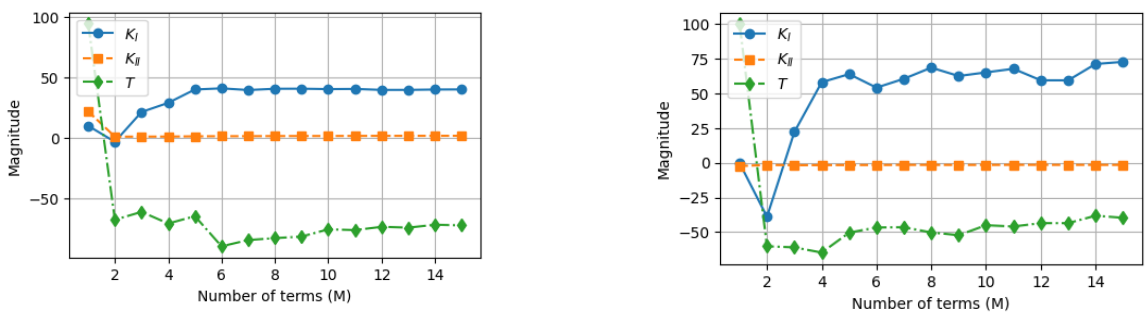

Fig 22 a) CT test & b)3PB test

As observed in the convergence plots, the values for KI, KII, and T-stress exhibit high volatility at low term counts but plateau and stabilize around M = 7 for both the CT and 3PB geometries. Consequently, M = 7 was selected to extract the final parameters for all subsequent analyses.

3.2 Displacement Field Verification

To verify the fidelity of the least-squares fit, theoretical displacement fields were reconstructed using the extracted parameters and overlaid onto the filtered experimental DIC data.

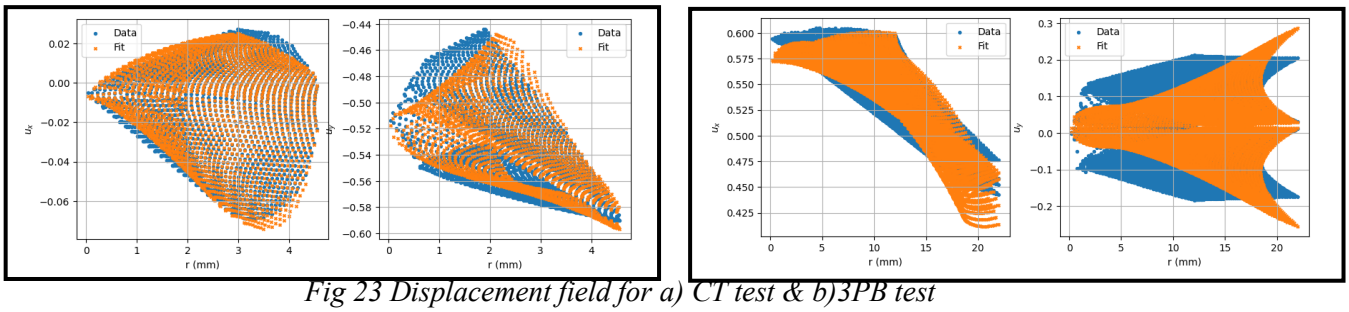

Fig 23 Displacement field for a) CT test & b)3PB test

The reconstructed analytical fields exhibit good spatial agreement with the DIC scatter data. Notably, the u_y (opening) displacements display the characteristic symmetric shape expected under pure Mode I loading.

3.3 Comparison of Fracture Parameters (K_I, K_{II}, and T-stress)

Table 1: Extracted fracture parameters for PMMA specimens.

Specimen Geometry	KI (MPa√m)	KII (MPa√m)	T-Stress (MPa)
CT	1.238	0.033	-84.793
3PB	1.911	-0.050	-46.471

The results yield several key mechanical insights:

- **Mode I Dominance:** Both the CT and 3PB specimens exhibit negligible sliding mode intensity (KII). This confirms that the experimental fixtures successfully maintained symmetric Mode I opening and that the numerical coordinate rotation successfully aligned the crack planes.
- **Applied Stress Intensity (KI):** While both specimens are composed of identical PMMA material (sharing the same critical fracture toughness KIc), the computed KI values differ (1.238 MPa√m vs. 1.911 MPa√m). This variance is expected, as KI represents the applied state of stress at the specific instant the DIC image was captured. The 3PB image frame corresponds to a higher applied remote load state than the CT frame.

- **T-stress and Geometry Dependence:** The T-stress is fundamentally geometry-dependent. Both configurations yielded negative T-stress values, indicating that an advancing crack would be directionally stable. The magnitude of the T-stress differs significantly (-84.79 vs -46.47 MPa) due to the distinct boundary conditions—pure bending in the 3PB specimen versus combined tension and bending in the CT specimen.

3.4 Verification of the Stress Singularity

To confirm the validity of the underlying LEFM assumptions, the analytical stresses directly ahead of the crack tip (theta=0) were analyzed to verify the square-root singularity predicted by Williams' theory.

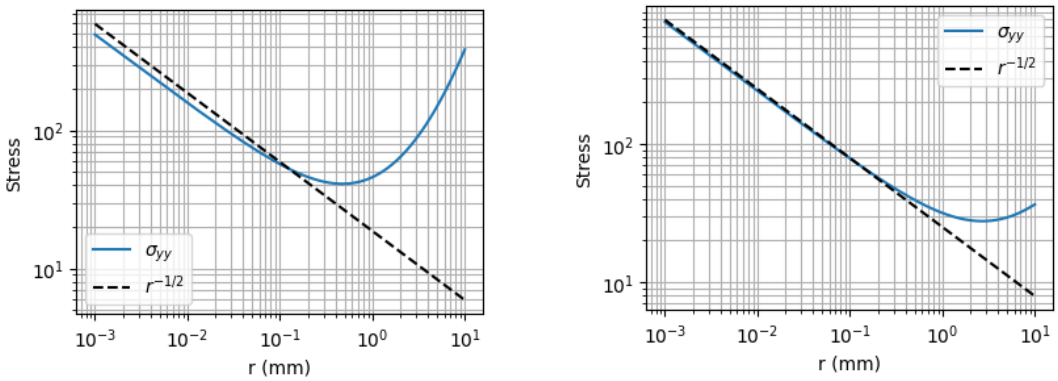

Fig 24 Log-log singularity check plots for a) CT test & b)3PB test

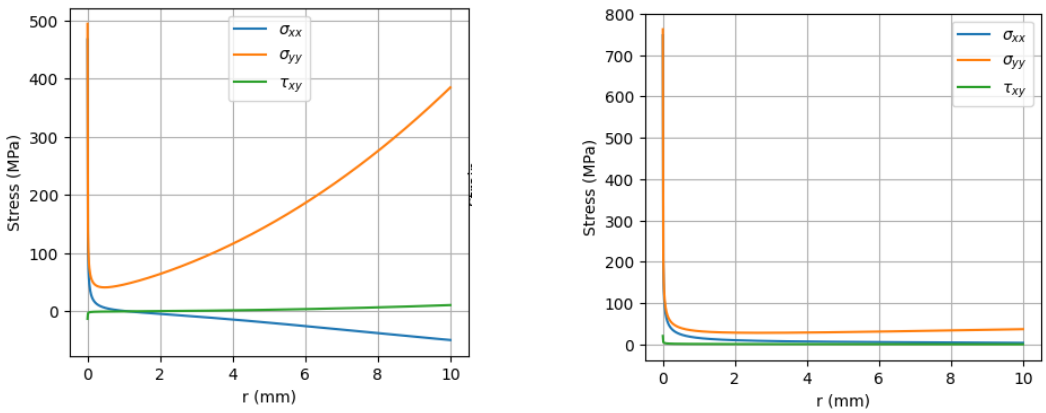

Fig 25 Stress field for a) CT test & b)3PB test

The log-log plots of the opening stress (σ_{yy}) versus radial distance parallel the theoretical reference line near the crack tip. This mathematically verifies the presence of the LEFM singularity and confirms that the DIC data was extracted from an appropriate annular region-far enough from the crack tip to avoid the non-linear cohesive zone, yet close enough to remain within the K-dominant region.

3.5 Sources of Uncertainty and Experimental Error

While the extracted fracture parameters demonstrated good convergence, the discrepancy in the calculated applied values between the identically composed PMMA specimens highlights the inherent sensitivities of optical experimental mechanics. Several potential sources of uncertainty and error contribute to this variance:

- **Manual Identification of the Crack Tip:** The analytical Williams expansion is highly sensitive to the spatial origin. Because the exact location of the crack tip was identified manually from the optical images during the coordinate transformation phase, a misalignment of even a few pixels significantly distorts the and coordinates. This spatial shift directly alters the displacement gradients, which propagates into the over-deterministic least-squares solver and skews the extracted and T-stress values.
- **DIC Region of Interest (ROI) Selection:** The data extraction relied on a Region of Interest (ROI) manually defined within the Ncorr software. Manual masking can inadvertently include noisy edge pixels near the crack opening or exclude valid, high-gradient data directly ahead of the crack tip. Because the least-squares fit weighs all unmasked data points equally, edge artifacts can drag the best-fit curve away from the true fracture parameters.
- **Subset Size and Image Noise:** Digital Image Correlation relies on tracking a random speckle pattern. Noise in the experimental data-stemming from uneven lighting, out-of-plane camera vibrations, or inconsistent speckle density-translates directly into displacement noise. Furthermore, the DIC subset radius acts as a spatial low-pass filter. If the subset size is too large, it effectively averages and "blurs" the steep displacement gradients near the singular crack tip, leading to an underestimation or shifting of the near-tip stresses.
- **Surface Measurements vs. Plane Strain Assumptions:** The Williams expansion implemented in this analysis assumes idealized 2D plane strain conditions. However, DIC strictly measures surface displacements. Because the free surfaces of a finite-thickness PMMA sheet exist in a state of plane stress, the optical surface data inherently captures 3D boundary effects that deviate slightly from the pure plane strain analytical model, introducing a systematic error in the resulting matrix calculation.

Discussion:

1. Accuracy of Stress Intensity Factor Extraction

The extracted Mode I stress intensity factor $K_I$was found to be dominant compared to K_{II}, indicating that crack opening mode governs the near-tip behavior. This is consistent with the applied loading and geometric symmetry of the problem. The small magnitude of K_{II}confirms minimal mode-mixity and validates the suitability of the displacement-based extraction method.

The inclusion of rigid body translation and rotation terms proved essential in achieving accurate parameter estimation. Without these terms, the least-squares solution would incorrectly attribute rigid motion effects to fracture parameters, leading to erroneous values of stress intensity factors. This observation aligns with findings reported in the reference study, where rigid body modes were shown to significantly influence displacement-based formulations.

2. Potential Sources of uncertainty and errors

Several factors may influence the accuracy of the computed fracture parameters:

- **Numerical Fitting Errors:** The least squares fitting approach is sensitive to the number of terms used in the Williams expansion. Using too few terms may lead to incomplete representation of the displacement field, while using too many terms may introduce overfitting.
- **Boundary Effects:** The remote boundary conditions applied in the finite element simulation were not explicitly known, which may introduce additional stress components not captured by the asymptotic formulation.
- **Higher-Order Effects:** The present study primarily focused on leading-order terms of Williams' series. Incorporating additional higher-order terms could improve accuracy but would require more data points and increased computational effort.

3. Convergence Behaviour in Williams Expansion

The convergence study demonstrated that the leading singular term converges rapidly, while higher-order terms contribute progressively less to the solution. The rapid stabilization of KI confirms that the crack tip field is primarily governed by the first Williams term.

In contrast, the convergence of T-stress required additional terms, reflecting its non-singular nature. This behaviour is consistent with theoretical expectations and previous literature, where higher-order terms are necessary to accurately capture constant stress contributions. Beyond a certain number of terms, no significant improvement was observed, indicating an optimal truncation order for stable and efficient computation.

Conclusion:

- The crack tip displacement and stress fields obtained from finite element data were analysed using Williams' asymptotic expansion to identify the crack opening modes for three different datasets.
- **Data-1** showed a dominant Mode I stress intensity factor with negligible Mode II contribution, indicating a **pure opening mode crack**. The symmetric displacement field and stable convergence of KI confirm this behaviour.
- **Data-2** exhibited significant non-zero values of both KI and KII, indicating a **mixed-mode crack condition** involving simultaneous opening and sliding. This was reflected in the asymmetric displacement fields and consistent convergence trends.
- **Data-3** demonstrated a dominant Mode II component, suggesting a **shear-driven crack behaviour**. The antisymmetric displacement patterns across the crack plane further supports this interpretation.
- For all datasets, the reconstructed displacement fields showed good agreement with finite element results, and the near-tip stress fields followed the expected $\mathbf{r^{-1/2}}$ **singularity**, validating the accuracy of the extracted fracture parameters. Overall, the methodology effectively distinguishes crack opening modes using displacement-based fracture analysis.
- The plane-strain SENB finite element model developed in Abaqus accurately captures Mode-I crack-tip behaviour under three-point bending, with stress and displacement fields showing physically consistent crack-tip concentration and symmetry.
- The extracted stress intensity factor $K_I$and T-stress evolve consistently with applied displacement, confirming correct implementation of loading, boundary conditions, and LEFM assumptions for the SENB configuration.
- The crack-tip stress singularity was numerically verified, with a fitted exponent $n \approx -0.4977$, in excellent agreement with the theoretical $r^{-1/2}$singularity predicted by Williams' asymptotic solution, thereby validating the fidelity of the finite element model.
- The multi-term Williams expansion methodology was successfully extended to process experimental Digital Image Correlation (DIC) data for physical PMMA Compact Tension (CT) and 3PB specimens. The extracted parameters confirmed pure Mode I fracture behavior for both geometries, while the variations in the calculated KI values correctly reflected the differences in the applied remote load states at the moment of image capture.
- Analysis of the experimental DIC displacement fields successfully verified the theoretical stress singularity near the crack tip for the physical PMMA specimens. Furthermore, the extraction of consistent negative T-stress values for both geometries mathematically confirmed the high constraint and directionally stable crack propagation expected during standard ASTM Mode I fracture testing.

Reference:

1] On the mode I fracture analysis of cracked Brazilian disc using a digital image correlation method (https://doi.org/10.1016/j.optlaseng.2015.10.006)

2] Quantifying crack tip displacement fields with DIC (https://doi.org/10.1016/j.engfracmech.2010.03.025)

3] Fracture Mechanics by C T Sun